D1: A* Search with Admissible Heuristic Design

FIT1061 Distinction Task — Week 5 — When Algorithms Choose Who Gets Help

Distinction tasks are where you begin to push beyond the standard approach and ask whether a more informed strategy can make search both more efficient and more principled. This task focuses on designing an admissible heuristic for A* so that the algorithm can use problem-specific information without sacrificing correctness.

Show us that you can

- You can explain what an admissible heuristic is and why it matters.
- You can compare A* with the earlier search methods.
- You can reason about the tradeoff between heuristic quality and search efficiency.

Give us

- A completed notebook with a implementation of A* with a heuristic.
- A short explanation of the heuristic and its effect on search behaviour.
- A reflection or comparison response.

Resources

- Week 3 lecture slides and recording *(Moodle)*
- D1 starter notebook *(OnTrack)*
- Week 3 Extra: A* Search (W03_astar_supplement.pdf) *(OnTrack + Moodle)*

Key sections in the task notebook

- D1.1: Implement A*
- D1.2: Heuristic design
- D1.3: Empirical comparison
- D1.4: Reflection
- Check your understanding

Do the following

1. Review the lecture material first.
2. Implement A* and test the heuristic.
3. Interpret the behaviour rather than just reporting scores.
4. Submit your completed work.

> **!** Acceptable use of Artificial Intelligence
>
> AI tools are **discouraged** for the core work at the Distinction level. You may use AI for *explanation* or *brainstorming*, but not to generate the core implementation, argument, or reflection. Cite all AI use. The task discussion will probe for genuine understanding.

All submitted work must be your own.